# Residual Penalties Versus Algebraic Completion In Conditional Diffusion

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## Abstract

Solar wind imputation involves uncertainty in both the mean magnetic field and its fluctuations. A variance identity links these statistics, while their probabilities depend on the available measurements. Conditional diffusion experiments examine how enforcing this relation affects predictive accuracy. A residual penalty improves the energy score at the end of training, but most of its mean advantage disappears after validation chooses the model. Generating fewer statistics and computing the remaining quantity algebraically gives the lowest mean score, with changes to both the representation and training loss. Its prediction intervals still miss too many measurements. For some baselines, clipping brings samples closer to the data but worsens the score because it reduces the ensemble spread term by more. A separate example uses an exact denoiser to show how a sampling approximation can produce estimates with small residuals while misrepresenting the conditional distribution. An algebraic restriction narrows the possible outcomes; estimating their probabilities remains a separate statistical problem.

## 1 Introduction

Missing solar wind measurements complicate estimates of magnetic turbulence [1]. Field vectors with large magnitudes can have a small mean when their directions differ. The same mean can accompany different fluctuations. Here, conditional diffusion infers the joint distribution of six magnetic statistics throughout a missing hour, using measurements before and after it. They describe the mean field, magnitude and variances in successive bins of five minutes.[1]

The variance identity expresses one statistic in terms of the other five. It follows from averaging vectors, independently of plasma dynamics, while many probability distributions can satisfy it. Applying the identity during training raises a further issue: a conditional mean can have a nonzero residual even when every possible realization satisfies the relation.

The experiments combine conditional diffusion [2] with the residual objective and sampling approximation of Physics-Informed Diffusion Models [3]. Earlier work examines residual minimization and distribution matching [4], or constructs outputs satisfying algebraic constraints [5]. Here, a penalty changes training, generating fewer coordinates changes the representation, and clipping transforms existing samples. Comparing these operations across training stages, and testing a sampler whose denoiser is available by calculation, shows how algebraic consistency and predictive accuracy can respond differently.

[1] Available at: https://github.com/magnaprog/diffusion-consistency

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## 2 Magnetic statistics and data

Let $B_1, \ldots, B_5$ be consecutive OMNI vector averages, each over one minute, from the same spacecraft in geocentric solar ecliptic coordinates. Define

$$\mu = \frac{1}{5}\sum_j B_j, \quad m = \frac{1}{5}\sum_j \|B_j\|, \quad v_s = \frac{1}{5}\sum_j (\|B_j\| - m)^2, \quad v_v = \frac{1}{5}\sum_j \|B_j - \mu\|^2. \quad (1)$$

The variances describe fluctuations in magnitude and in the vector, respectively, among the minute averages. The six variables are $z = (\mu_x, \mu_y, \mu_z, m, v_s, v_v)$. Expanding both population variances gives

$$R_{\text{obs}}(z) = m^2 + v_s - \|\mu\|^2 - v_v = 0. \quad (2)$$

Omitting the variances gives $R_{\text{inst}} = m^2 - \|\mu\|^2$. For realizable statistics, it measures directional cancellation, weighted by field amplitude. It vanishes when all nonzero vectors share a direction, even if their magnitudes differ. Reducing amplitudes while retaining their directions can also lower this residual.

Write $r = \|\mu\|$. Five vectors in three dimensions, including possible zero vectors, can realize these statistics exactly when

$$m \geq r, \qquad 0 \leq v_s \leq \frac{3}{2}m^2 + \frac{5}{2}r^2, \qquad v_v = m^2 + v_s - r^2. \quad (3)$$

Appendix A proves sufficiency. The equality and inequalities play different roles. Deriving $v_v$ enforces the equality while preserving the other five variables and their primary score. Clipping also brings $m$ and $v_s$ into the ranges in Equation 3, which can change that score.

The same identity behaves differently when applied to a conditional mean. For conditioning information $Y$,

$$R_{\text{obs}}(\mathbb{E}[z \mid Y]) = \operatorname{tr}\operatorname{Cov}(\mu \mid Y) - \operatorname{Var}(m \mid Y). \quad (4)$$

Either sign is possible, even when every possible sample satisfies the identity. Thus, a nonzero residual at the denoiser's mean prediction can arise from conditional uncertainty. Evaluating residuals at individual samples instead checks the algebraic relation in each possible realization [3].

**Data construction.** OMNI combines solar wind measurements from several spacecraft [6]. The inputs here come from its modified high resolution product [7], whose timestamps estimate arrival at the bow shock nose [8]. Propagation and averaging precede masking, so the reconstruction concerns this arrival-time record. Magnitudes are Euclidean norms of the vector components. Each window spans four hours in 48 bins of five minutes. Masking the central twelve bins leaves 90 minutes of conditioning data on each side. Complete windows use one spacecraft within one month, retaining 12.77% of possible training starts.

The study covers twenty years, 2005 through 2024, with separate training, test and validation periods within each year and unused dates between them. Normalization uses distinct training bins. Evaluation uses windows without overlap and limits their number per month, giving 319 validation and 480 test windows. Appendix B gives the calendar periods and sampling procedure.

## 3 Models and evaluation

**Training.** D uses diffusion to generate all six variables [9]. I-S adds a penalty on $R_{\text{inst}}$; O-S adds one on the variance identity $R_{\text{obs}}$. G5 generates five variables and derives $v_v$ after sampling. All four models receive all six variables in the available intervals. Training and sampling add noise only within the missing hour and preserve the available measurements.

For standardized state $u$ in the missing hour, the training objective adapts Bastek et al. [3] by combining mean squared error, weighted by Min-SNR-5 [10], with a residual term:

$$\mathcal{L}_t = w_t \ \text{MSE}(\hat{u}_0, u_0) + \frac{c_R}{2\Sigma_t}\,\frac{1}{12}\sum_{k=1}^{12}\left(\frac{R(z^*_{0,k})}{B^2_{\text{ref}}}\right)^2. \quad (5)$$

Here $\hat{u}_0$ estimates the original signal. Two denoiser evaluations give $z^*_{0,k}$ in physical units, except at $t = 1$, where the first estimate is reused. The data weight is $w_t = \min(\text{SNR}_t, 5)$; $\Sigma_t$ is the diffusion

posterior variance, with $\Sigma_1 := \Sigma_2$. The scale $B^2_{\mathrm{ref}}$ averages $m^2 + v_s$ over distinct training bins. A cosine schedule uses 100 steps [11]. D and G5 use $c_R = 0$; I-S and O-S use $c_R = 0.0005$.

Each method uses three training seeds, with model weights saved at five stages. These states, or checkpoints, have the same update counts across methods (Appendix C). The penalty adds denoiser evaluations and computational cost. The test period had been examined during earlier development; the comparison retains the original settings and penalty coefficient without further tuning on that period.

The energy score evaluates the joint predictive distribution through Euclidean distances; lower scores are better [12]. The primary score uses $(\mu_x, \mu_y, \mu_z, m, v_s)$ at twelve hidden positions after standardization using training data. For $K$ independent draws from a fixed predictive distribution, an unbiased estimate subtracts half the mean distance between distinct draws from their mean distance to the withheld values. The pairwise mean uses all $K(K-1)$ ordered pairs. The two terms measure proximity to observations and separation within the ensemble. Monthly means receive equal weight. Continuous ranked probability scores, interval coverage and width assess individual coordinates (Appendix C). Diffusion results average three training seeds, with coverage calculated separately for each ensemble. Intervals for score differences are calculated by resampling years or consecutive groups of two years with replacement, using the same resampled years for both methods. They measure temporal variation for the existing fits and samples. Grouping test cases by withheld outcomes uses information available only after the missing values are revealed.

**Model selection and baselines.** Validation compares ensembles of 39 samples from 25 denoising diffusion implicit model (DDIM) evaluations with $\eta = 0$ [13]. It minimizes the energy score with equal month weights, choosing checkpoints separately before and after clipping; ties favor the earlier state. Test ensembles use 39 samples and 50 evaluations.

Linear interpolation connects the measurements immediately before and after the missing hour; persistence repeats the preceding bin. Gaussian interpolation adds a Gaussian residual fitted on training data, with covariance shrinkage $\rho = 0$. Gaussian regression uses the full available context to predict the residual mean and uses the same residual covariance for every window, then derives $v_v$. Validation chooses its ridge and covariance shrinkage parameters (Appendix G).

Table 1: Results for 480 test windows with equal month weights. Energy scores use the 60 standardized primary coordinates. Diffusion results average three fits; each other baseline uses one ensemble realization. Validation selects checkpoints separately for raw and clipped diffusion predictions. Coverage is averaged over five primary variables at twelve hidden positions. $|R|$ is the mean absolute value of $R_{\mathrm{obs}}$ in $\mathrm{nT}^2$. The last column gives the percentage of bins that fail a moment inequality beyond rounding, excluding the equality test. Interpolation and persistence have intervals of zero width.

| | Energy score | | 90% coverage (%) | | Raw samples | |
|---|---|---|---|---|---|---|
| Procedure | Raw | Clipped | Raw | Clipped | $\lvert R\rvert$ | Bins failing inequalities (%) |
| D | 1.888 | 1.869 | 72.8 | 71.0 | 2.956 | 50.4 |
| I-S | 1.882 | 1.876 | 69.4 | 69.6 | 3.732 | 37.8 |
| O-S | 1.885 | 1.866 | 70.0 | 71.1 | 2.680 | 55.7 |
| G5 | 1.785 | 1.784 | 75.7 | 75.8 | 0.000 | 48.8 |
| Interpolation | 2.393 | 2.393 | 0.0 | 0.0 | 3.459 | 0.0 |
| Persistence | 3.067 | 3.067 | 0.1 | 0.1 | 0.000 | 0.0 |
| Gaussian interpolation | 2.125 | 2.229 | 94.3 | 93.6 | 15.491 | 77.7 |
| Gaussian regression | 1.935 | 2.035 | 94.0 | 93.5 | 0.000 | 74.6 |

## 4 Results and interpretation

The penalty comparison depends on when the models are evaluated (Table 1, Figure 1a). At the final update, O-S with the original coefficient has a lower raw energy score than D for all three seeds.

Validation chooses earlier D checkpoints, leaving little mean difference: O-S minus D is $-0.0028$, with a 95% interval of $[-0.0346, 0.0242]$ from resampling years. Relative intervals from both resampling schemes remain within the original $\pm 2\%$ comparison range for these three fits.

G5 has the lowest mean primary score: O-S minus G5 is 0.1002, with interval $[0.0712, 0.1391]$ from resampling years. Both the O-S/D and O-S/G5 comparisons reverse sign across seeds. G5 changes the representation and training loss together, so the comparison includes both effects. Algebraic completion alone would preserve the five primary quantities and their score for the same samples. Most of G5's mean improvement over D appears in the spread term, with different contributions at each seed (Figure 4). Its intervals still cover too few measurements, especially for variance in field magnitude (Figure 1b), and nearly half of its raw bins fail the moment inequalities.

The clipping comparison transforms each sample in an existing ensemble (Table 13). Model weights and evaluation windows stay the same. It moves G5's samples with little change in their mean energy score. For Gaussian interpolation, it brings samples closer to the withheld measurements, but the subtracted spread term falls by more, increasing the score. D's change in Table 1 also includes a different checkpoint chosen after clipping (Figure 5).

The hidden Markov example examines sampling without error from fitting the denoiser (Figure 3). Its exact denoiser averages over the unknown switching probability. New hidden paths and independent forward noise give a distribution of estimates for each available context. Across 36 cases at $t = 100$, two evaluations give estimates with mean absolute $R_{\text{obs}} = 4.17 \times 10^{-5}$ in HMM units, but energy score 6.958 versus 4.164 for independent conditional samples. Separately, DDIM sampling on 288 windows gives a poor score with two evaluations and scores close to the conditional reference with 25, 50, or 100 (Table 2). All scores have finite sampling error.

## 5 Conclusion

Moment consistency is a property of each magnetic realization; predictive accuracy also concerns how probability is distributed among realizations. The comparisons show how this distinction enters training, representation and sampling. Proper scores and interval coverage quantify differences that remain after the algebraic relation is satisfied. The empirical ranking concerns three seeds and hours masked in complete records from a test period examined earlier. Naturally missing records, later years and other interval lengths would test how far it extends.

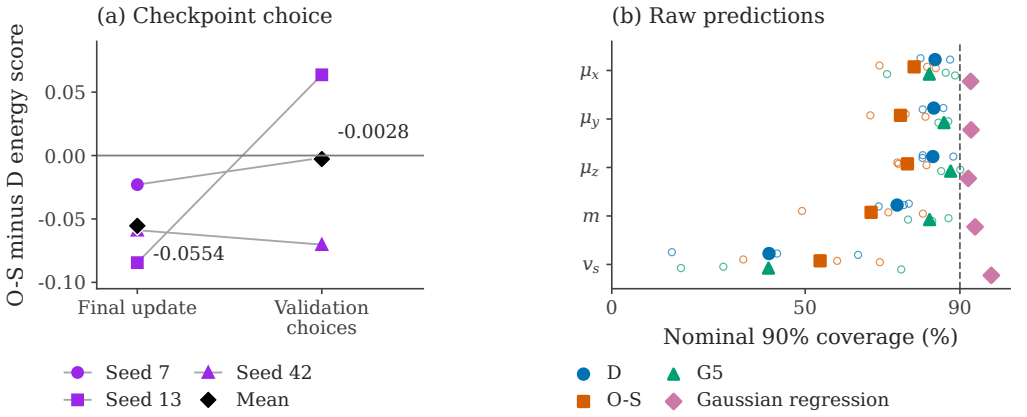



Figure 1: Model choice and predictive uncertainty. (a) O-S minus D energy score at the final update and at the checkpoints chosen independently by raw validation score. Symbols distinguish the three training seeds; diamonds show their mean. Negative values favor O-S. Validation with 25 or 50 evaluations chooses the same six checkpoints. (b) Raw nominal 90% interval coverage for each magnetic quantity. Large filled markers average three diffusion fits; small open circles show individual fits. Gaussian regression uses one ensemble per window. The dashed line marks 90%. Scores and coverage give equal weight to months; coverage also averages the twelve hidden positions.

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## A Realizability of magnetic moments and energy

The upper bound on $v_s$ follows from the lengths of the five vectors. Let $a_j = \|B_j\|$, with total length $\sum_j a_j = 5m$. Their sum can have length $5r$ precisely when $\max(0, 2\max_j a_j - 5m) \le 5r \le 5m$. Hence, each length is at most $5(m + r)/2$. With the total fixed, the sum of squared lengths is largest when two lengths are $5(m + r)/2$ and $5(m - r)/2$ and the other three are zero. Taking these two vectors as opposite multiples of one unit vector gives the required resultant. Therefore, $m^2 + v_s \le 5(m^2 + r^2)/2$, which gives the upper bound in Equation 3.

Equal lengths give $v_s = 0$, while the lengths above attain its upper bound. The nonnegative length tuples with sum $5m$ and individual upper limit $5(m + r)/2$ form a convex set containing both tuples. Along the segment between them, the sum of squares varies continuously, attaining every intermediate variance. Each tuple still admits a vector sum of length $5r$. Rotating all five vectors together aligns a nonzero sum with $\mu$; a zero sum already has the required mean. This proves sufficiency, including the case in which all five vectors vanish.

The common quantity $Q = m^2 + v_s = \|\mu\|^2 + v_v$ is the mean squared magnitude of the five vectors, each an average over one minute. Multiplication by $10^{-18}/(2\mu_0)$ expresses it as the magnetic energy density represented by those averages, after converting $\mathrm{nT}^2$ to $\mathrm{T}^2$. Variation inside the original minute averages is absent from this quantity.

Averaging $Q$ across consecutive bins preserves the mean squared magnitude over the combined interval. This property also holds numerically for all 97,699 eligible intervals of 15 minutes that begin on a UTC quarter hour within the union of training windows: the calculation from three bins differs from that using the original fifteen vectors by at most $9.09 \times 10^{-13}\,\mathrm{nT}^2$. Replacing each bin by its mean vector discards its vector variance and reduces the result by $0.8943\,\mathrm{nT}^2$ on average.

## B OMNI measurements and evaluation sample

The magnetic inputs are the vector components in geocentric solar ecliptic coordinates, GSE. Their norms define the magnitudes used here, which can differ from NASA's separately supplied average field magnitude. Before bins of five minutes are formed, data checks exclude nonfinite entries and fill values from the OMNI specification. They verify integer calendar fields, unique timestamps in chronological order and one spacecraft throughout each window. The study applies no additional amplitude threshold or removal of spikes according to the values subsequently withheld.

OMNI first estimates arrival times and forms weighted magnetic averages for each minute. Neighboring output minutes can share underlying measurements, and propagation normals depend on nearby magnetic observations. The study masks the resulting record, retaining any influence of the hidden interval on those earlier estimates. NASA also keeps data whose predicted arrival order differs from their order at the spacecraft and describes the resulting mixing of plasma regions [8, Sections 3a–3c and Appendix 5]. Every window uses one spacecraft, while propagation can still change the ordering of its records. Cancellation between field directions, variance and changes at the edges of a missing interval can reflect both this processing and solar wind variability. The prediction intervals describe uncertainty about the withheld processed values, treating the instrument measurements and arrival-time estimates as given.

Which intervals remain complete depends on propagation as well as solar wind conditions. Magnetic records may be omitted when suitable plasma velocities are missing or when a long interval with little field variation gives no acceptable phase front normal. Consequently, completeness is an ambiguous indicator of quiet or disturbed solar wind. NASA chooses among spacecraft in segments of three hours using coverage, spatial separation and continuity of source, with a preference for Wind from 2020 onward [8, Appendix 6]. Thus, comparisons between spacecraft combine differences in instruments, dates and processing. The experiments use the original downloaded OMNI records.

The duration between observing times, DBOT, is the difference between the inferred spacecraft observing times of successive records on the OMNI arrival-time grid. A negative value indicates an apparent reversal between these two orderings. Across the training dates, negative DBOT occurs in 555,719 of the 4,298,363 valid magnetic minutes with known DBOT, and in at least one minute of 55,339 of the 300,392 eligible bins. The phase front normal interpolation field measures the

percentage of contributions to a minute magnetic average whose propagation normal was interpolated [7]. These minute counts include valid magnetic records outside eligible training windows.

The completeness criteria retain 122,309 of the 957,720 possible training starts. A possible start lies on the grid of five minutes and places all four hours inside a calendar month assigned to training. The retained windows contain 163,988 distinct bins that can occur in the missing interval. Availability varies between years, with 11,340 starts in 2005 and 2,568 in 2014. The contribution of solar activity to these differences has not been established.

In each year from 2005 through 2024, training uses 15 January through 30 June, test uses August and September, and validation uses 1 November through 17 December. Unused dates separate these periods. Within each evaluation month, complete, nonoverlapping windows of four hours are formed in chronological order. From $N$ windows, $q = \min(N, q_0)$ are retained at indices $\lfloor (j + 0.5)N/q \rfloor$ for $j = 0, \ldots, q - 1$, with $q_0 = 8$ for VAL and $q_0 = 12$ for TEST. The indices spread the evaluation windows across this sequence, and the quota limits differences in monthly sample counts. Eligibility still depends on completeness.

## C Diffusion training and evaluation

The four methods each have three fits, allowing comparisons across training seeds at the same update counts. Each fit receives 29,636 optimizer updates with batches of 128, using one batch gradient per update. This corresponds to 31 passes through the training data, including the shorter batch at the end of each pass.

The U-Net has channel widths 64/128/128 and contains 1,603,398 parameters when it generates six coordinates, or 1,603,013 for G5. The architecture omits attention and dropout. Evaluation uses the fitted weights directly, without exponential averaging. AdamW uses learning rate $2 \times 10^{-4}$, betas $(0.9, 0.999)$, epsilon $10^{-8}$, weight decay $10^{-4}$, and a limit of 1 on the norm of the total gradient.

Model states are saved after 2,868, 9,560, 16,252, 22,944, and 29,636 optimizer updates. These checkpoints provide common stages of training for both validation and comparison on TEST.

The cosine diffusion schedule has offset 0.008 and an upper limit of 0.999 for beta. Its posterior variance is $\Sigma_t = \beta_t(1 - \bar{\alpha}_{t-1})/(1 - \bar{\alpha}_t)$, with $\Sigma_1 := \Sigma_2$. The measurements and score calculations use double precision, float64; network calculations and diffusion samples use single precision, float32. PCG64 generates the initial evaluation noise for six coordinates, of which G5 uses the first five. A CPU PyTorch generator supplies training timesteps and Gaussian noise. Every denoiser evaluation receives the same available measurements.

The energy score also evaluates either the three vector components or all six magnetic quantities at the hidden positions. The continuous ranked probability score (CRPS), interval coverage and width assess scalar predictive distributions. Identity residuals and inequality failure rates assess moment consistency, while errors at the gap edges compare predicted changes with those in the withheld measurements.

G5 changes both what the network generates and how its error is averaged. Its data loss averages squared error over 60 hidden coordinates, compared with 72 for the other models. Removing the $v_v$ prediction changes the weight of each remaining coordinate in that average and the network's input and output dimensions. These changes enter the comparison together. Deriving $v_v$ from an existing ensemble preserves the five primary coordinates and their score.

Prediction intervals are formed for each coordinate by sorting its 39 samples. The second and thirty-eighth values enclose the nominal 90% interval; the tenth and thirtieth enclose the 50% interval. Both ends are included. If the withheld value and ensemble members are independent samples from the same continuous conditional distribution, rank symmetry gives expected coverage $(h - l)/40$ between ranks $l$ and $h$. Ties in the discrete HMM distribution can increase this coverage.

Errors at gap edges compare predicted and observed changes between hidden and available bins, separately from changes inside the gap. For interpolation, the mean-vector changes at both edges are $(\mu_{30} - \mu_{17})/13$, which can be nonzero. The Gaussian model also permits departures from the interpolant at those edges. The identity residual measures departure from the equality; the fractions in the tables count failures of the physical inequalities.

The residual estimators use different approximations to the original data. M evaluates the residual at the first denoiser prediction. S takes a DDIM step from $t$ to 1 without adding noise, then predicts the original data at that lower noise level. Both predictions return to physical units through the inverse affine transform before their residuals are evaluated. At $t = 1$, S reuses the first prediction and coincides with M. For a fixed noisy input, S is deterministic; it is used as an approximation to a conditional sample. Appendix D tests the distribution of these estimates using an exact denoiser.

To examine how the penalty enters optimization, the gradient comparison evaluates the weighted data and residual terms on the same batches before clipping the total gradient. For each seed, the same eight TRAIN and eight VAL windows are used at the first and final O-S checkpoints, with noise seed 7003 and $c_R = 0.0005$. Both residuals and both estimators are evaluated at $t \in \{1, 25, 50, 100\}$ without changing the model weights. Norms and angles describe the relative size and direction of these gradients; undefined ratios and angles remain missing.

### C.1 How the residual penalty is weighted

A constant $c_R$ still gives different relative penalty weights across noise levels. With $\lambda_t = c_R/(2\Sigma_t)$ and $c_R = 0.0005$, the ratios $\lambda_t/w_t$ at $t = 1, 25, 50, 100$ are 0.1239, 0.004183, 0.008642, and 1030.7. The large final ratio accompanies a Min-SNR data weight of $2.429 \times 10^{-7}$. To understand the effect on optimization, the derivatives of both loss terms also matter.

At the final O-S checkpoints, the ratio of residual to data gradient norms ranges from 315 to 1799 on TRAIN and 214 to 1266 on VAL at $t = 100$. At $t = 25$, the ranges are 0.0081 to 0.0267 and 0.0103 to 0.0465. Each range spans three seeds, with eight windows per seed. In absolute terms, however, the TRAIN residual-gradient norm at $t = 100$ is only $7.379 \times 10^{-5}$ to $4.401 \times 10^{-4}$, compared with 0.1396 to 0.4587 at $t = 1$. Thus, a large ratio can accompany a small gradient. Optimizer updates combine noise levels and clip the total gradient afterward. These ratios describe the loss gradients at individual noise levels; their effect on prediction quality also depends on the rest of the training trajectory.

### C.2 Choosing checkpoints and prior use of test data

All models were refitted after revisions to the evaluation. The sampler now supplies the same available measurements during training and inference, and the HMM calculation averages over the switching probability that is unknown to the prediction methods. These revisions change both the fits and their evaluation, so differences from earlier scores cannot be attributed to either change separately. The test period had already been examined before this comparison.

D at seed 7 is the only fit for which clipping changes the checkpoint chosen by validation: raw predictions choose 9,560 updates, while clipped predictions choose 29,636. For D, the mean energy score across the three training seeds is 1.881 when clipping uses checkpoints chosen from raw predictions, and 1.869 when each checkpoint is chosen by its clipped score.

The original superiority criterion compares raw O-S with raw D, raw G5, and D and G5 after clipping with independently chosen validation checkpoints. O-S meets the criterion only if its mean energy score is at least two percent lower in every comparison, both difference intervals lie below zero, and every seed favors O-S. A separate criterion compares raw O-S with raw D using a prescribed tolerance: both relative intervals lie strictly inside $[-0.02, 0.02]$, with positive D score denominators in every resample. These criteria assess the size and consistency of score differences, without implying equality of predictive distributions.

For O-S minus D, resampling groups of two consecutive years gives an interval of $[-0.0345, 0.0233]$. The intervals describe temporal variation separately for each comparison, conditional on the existing fits and draws. They omit adjustment for multiple comparisons and variation from new training seeds.

### C.3 Training stage and checkpoint choice

Validation originally used fewer denoiser evaluations than TEST. The checkpoint analysis asks whether that difference affects model choice. It evaluates all five states of D and O-S at each training seed, using the original weights, windows, 39 samples and evaluation seed 7. Thirty validation ensembles with 50 evaluations reuse the initial noise from the original ensembles with 25. Their

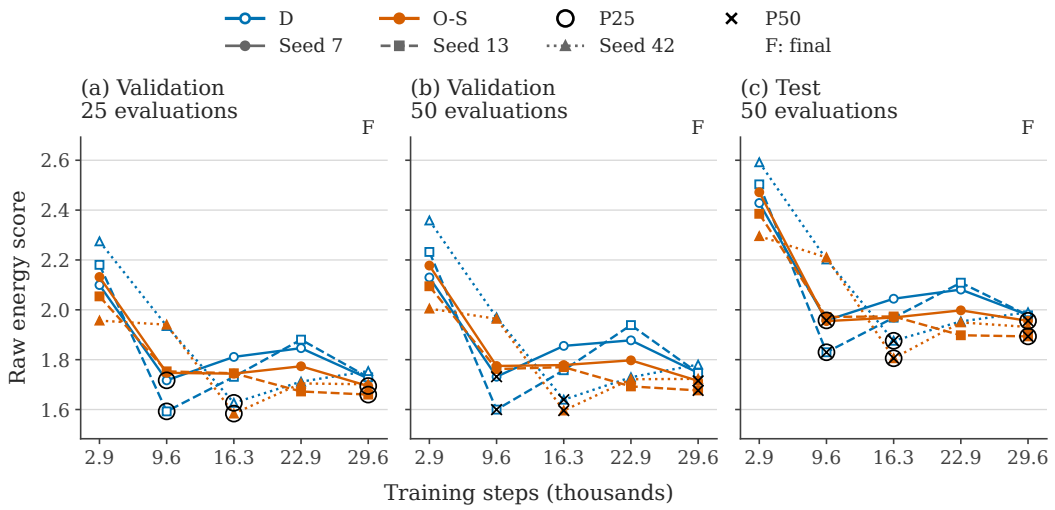



Figure 2: How the D/O-S comparison changes during training. The subplots show VAL with 25 DDIM evaluations, VAL with 50, and TEST with 50. Each point gives equal weight to months; ensembles contain 39 samples. Colors and open or filled markers distinguish D and O-S, while line styles and marker shapes distinguish seeds. Black rings mark the validation choice with 25 evaluations, P25; crosses mark the choice with 50, P50. Both choices coincide for all six fits. F marks the common final update. Validation and test scores concern different windows. The P25 choices were fixed before this additional TEST comparison; the TEST period had been examined during earlier development.

validation scores determine the model choices before additional TEST samples are generated. TEST then evaluates all thirty model states with 50 evaluations (Figure 2). This analysis still concerns the test period examined earlier.

All six validation choices agree between 25 and 50 evaluations for these fits and initial noise. The penalty comparison nevertheless changes across training: at the five common update counts, mean TEST differences for O-S minus D are $-0.1238$, $0.0490$, $-0.0470$, $-0.0993$, and $-0.0554$. O-S has lower scores at all three seeds at the last two counts; seed differences have mixed signs at the first three.

Validation chooses an earlier D checkpoint for every seed, but the final O-S checkpoint for seeds 7 and 13. Moving all fits to the final update raises D's mean TEST score by 0.0945 and O-S's by 0.0419, changing O-S minus D from $-0.0028$ to $-0.0554$. The larger increase for D explains the arithmetic change in their difference. Establishing whether regularization or overfitting causes this pattern is a separate empirical question.

Model choice can also depend on which years appear in validation. In 10,000 resamples of the twenty years, D at seed 7 chooses update 9,560 in 61.0% of resamples with 25 evaluations and 71.5% with 50; the other resamples choose the final update. D at seeds 13 and 42 retains its respective earlier choice in every resample under either evaluation count. Each sampled year contributes both validation months, including repeated occurrences, and all checkpoints use the same resampled years. With weights and samples held constant, these frequencies measure sensitivity to the years represented in validation.

Repeating checkpoint choice also tests whether this sensitivity changes the O-S/D comparison. In 10,000 resamples at seed 7003, the same year multiplicities weight validation and TEST. Validation with 50 evaluations chooses each fit's checkpoint anew, with ties favoring the earlier update. The 2.5th and 97.5th percentiles of O-S minus D are $[-0.0336, 0.0246]$, close to $[-0.0346, 0.0242]$ when the original choices are retained in the same resamples. Thus, allowing model choice to vary preserves the small aggregate difference in this exercise. Both calculations condition on the existing fits and samples; uncertainty from retraining and nominal coverage under temporal dependence remain outside this sensitivity check.

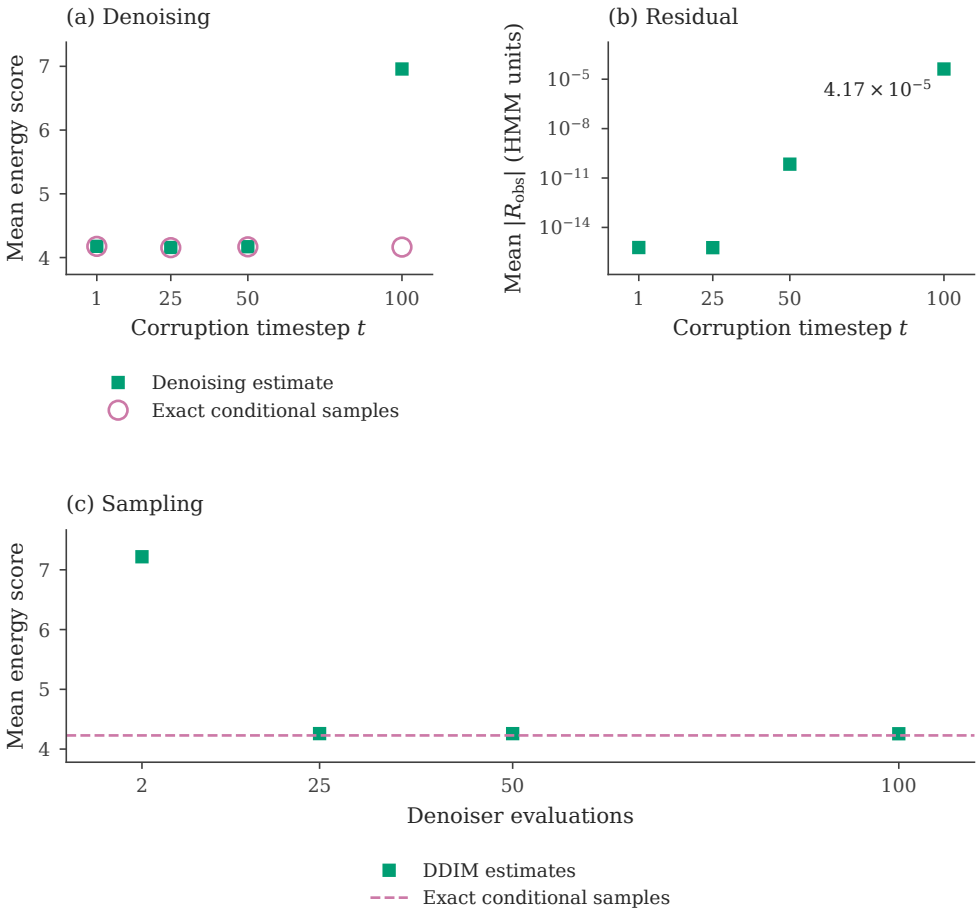



Figure 3: The stationary hidden Markov model separates residual size from distributional accuracy. Subplots (a,b) use the same 36 cases at each timestep, each with 128 conditional paths and independent noise. (a) Squares give the mean energy score of denoising estimates; open circles estimate the expected score of the conditional distribution using an independent random stream. (b) Mean absolute residual on a logarithmic axis. The first two means are near floating point precision. The estimator uses one denoiser evaluation at $t = 1$ and two at the other timesteps. (c) Complete DDIM sampling uses 39 samples on each of 288 windows, a separate evaluation population. The dashed line gives the score of independent samples from the exact conditional distribution. All scores are finite Monte Carlo estimates.

## D A stationary hidden Markov model with a known distribution

This example makes the conditional distribution available by calculation, so denoising and sampling can be examined without error from fitting a network. The two states, $z_0$ and $z_1$, are magnetic statistics that obey the same identity as the solar wind data. For $\alpha \in \{0, 1, 2\}$, define $c_j = j - 3$ for $j = 1, \ldots, 5$ and $d_\alpha = \sqrt{1 + 2\alpha^2}$. The vectors that produce the two states are

$$B_j^{(0)} = \frac{(1, \alpha c_j, 0)}{d_\alpha}, \qquad B_j^{(1)} = \frac{2(1 + 2\alpha c_j,\ 2 - 2\alpha c_j,\ 2 + \alpha c_j)}{3 d_\alpha}. \tag{6}$$

The mean squared field magnitudes are $Q_0 = 1$ and $Q_1 = 4$ in HMM units for every $\alpha$. Thus, the geometry and magnitude variance can change while each state retains its own value of $Q$, preserving the difference between the states. Standardization gives equal weight to the six states and uses $B_{\text{ref}}^2 = 2.5$; these values are set before evaluation.

A two-state Markov chain determines the sequence, with switching probability $p \in \{0.1, 0.25, 0.5\}$ and transition matrix $T_p = \left(\begin{smallmatrix} 1-p & p \\ p & 1-p \end{smallmatrix}\right)$. Twelve hidden states connect the left state $a$ to the right state $b$, giving thirteen transitions. Stationarity sets $\Pr(a) = 1/2$ and $\Pr(b \mid a, p) = (T_p^{13})_{ab}$. The hidden sequence is then sampled conditional on these two states.

Each record has 48 bins spanning four hours. The 18 bins before the missing hour repeat the left state's statistics, and the 18 bins after it repeat the right state's statistics. This context reveals $\alpha$ and the two end states, while the switching probability $p$ remains unknown. The twelve intervening bins follow a Markov trajectory conditional on the end states.

Inference averages over the three possible switching probabilities, each with prior probability $1/3$. The end states give weights proportional to $\Pr(p)(T_p^{13})_{ab}$. At each denoiser evaluation, the likelihood of the current noisy input further updates these weights. The calculation and the prediction methods have the same available information.

For posterior probability $q$ of state 1, define $\kappa_\alpha = [1 + 2\sqrt{1+\alpha^2} + 2\sqrt{1+4\alpha^2}]/5$. The posterior mean has residual

$$R_{\mathrm{obs}}\big((1-q)z_0 + qz_1\big) = q(1-q)\frac{11/3 - \kappa_\alpha^2}{1+2\alpha^2}. \tag{7}$$

The coefficient is positive at $\alpha = 0, 1$ and negative at $\alpha = 2$. Both states satisfy the identity, yet their posterior mean has a nonzero residual whenever $0 < q < 1$. A squared residual penalty assigns a positive loss to this exact posterior mean. This magnetic example illustrates the nonlinear expectation problem discussed by Bastek et al. [3].

Evaluation uses 32 paths for each switching probability. Each path is reused at the three values of $\alpha$, giving 288 windows and allowing sampling accuracy to be compared across geometries with the hidden state sequence held fixed.

Table 2 compares sampling methods on these 288 windows, using stationary end states, the same available information and 39 samples per method. The approximate samplers start from the same noise. Independent samples from the exact conditional distribution provide the first row: this distribution minimizes the expected energy score, while the finite estimate in the table varies with Monte Carlo error. DDIM with two denoiser evaluations has a substantially higher score; the scores with 25, 50 and 100 evaluations lie close to the estimate for the exact distribution. The earlier design with independent end states forms a separate sensitivity analysis.

Table 2: Primary energy scores for 288 stationary hidden Markov windows. The exact conditional distribution averages over the unknown switching probability. DDIM uses the exact denoiser with the evaluation counts shown. All scores come from finite ensembles.

| Procedure | Energy score |
|---|---|
| Exact conditional draws | 4.2287 |
| Two evaluations of exact denoiser | 7.2183 |
| Exact denoiser, 25 evaluations | 4.2591 |
| Exact denoiser, 50 evaluations | 4.2586 |
| Exact denoiser, 100 evaluations | 4.2567 |

The denoising comparison distinguishes two conditioning sets. For one fixed noisy input, it examines the estimate and the posterior given that input and the available context. Figure 3(a) instead holds only the end states and vector geometry fixed: each estimate starts from a new hidden path and independent forward noise, and its distribution is compared with independent paths from the conditional distribution for that same context. The reference uses a separate random stream. Each calculation covers 36 cases at each of four timesteps, giving 144 cases in total. A posterior concentrated on one path at low noise can be correct for that noisy input. The separate comparison across repeated paths and noise assesses the distribution given the available context.

### D.1 Training diffusion models on the same HMM

The learned models test how these residual objectives behave during training. Each fit uses 4,096 windows generated at seed 106 by cycling through the nine $(\alpha, p)$ combinations, a U-Net with channel widths 32/64/64, batches of 128, and 5,000 optimizer updates at seed 7. The remaining

optimizer and diffusion settings follow Appendix C. Table 3 compares nine fits to the stationary chain with 50 DDIM evaluations for sampling. M and S denote the mean estimate and the estimate from two denoiser evaluations defined in Appendix C. At both coefficients, the identity penalty gives primary scores close to D, whereas omitting the variances gives higher scores with either estimator. These comparisons concern one training seed and its existing samples; variation across training seeds remains outside this comparison. The earlier models with independent end states belong to a separate analysis.

Table 3: Nine diffusion fits to the stationary hidden Markov model after 5,000 optimizer updates at seed 7. Evaluation uses 288 windows and 39 samples per window; clipping acts on these same ensembles. Raw $|R|$ is the mean absolute residual of the observation identity in HMM units. The inequality column gives the percentage of generated bins that fail a moment inequality beyond rounding tolerance, separately from the equality residual. Coverage averages the five primary coordinates and twelve positions. The independent exact conditional ensemble has 100% coverage for the inclusive nominal 90% intervals.

| | | Energy score | | | Raw samples | |
|---|---|---|---|---|---|---|
| Procedure | $c_R$ | Raw | Clipped | $\lvert R\rvert$ | Bins failing inequalities (%) | 90% coverage (%) |
| D | 0 | 4.2607 | 4.2609 | 0.0126 | 27.5 | 94.5 |
| I-M | 0.0005 | 4.3283 | 4.3285 | 0.2861 | 26.6 | 84.3 |
| I-M | 0.005 | 4.6021 | 4.6022 | 0.7050 | 26.6 | 81.2 |
| I-S | 0.0005 | 4.4991 | 4.4993 | 0.6262 | 26.0 | 77.5 |
| I-S | 0.005 | 4.8020 | 4.8020 | 1.0086 | 29.2 | 74.6 |
| O-M | 0.0005 | 4.2604 | 4.2605 | 0.0122 | 27.8 | 94.7 |
| O-M | 0.005 | 4.2544 | 4.2545 | 0.0095 | 27.4 | 95.3 |
| O-S | 0.0005 | 4.2593 | 4.2595 | 0.0120 | 27.8 | 94.8 |
| O-S | 0.005 | 4.2551 | 4.2553 | 0.0094 | 27.4 | 95.9 |

The estimate used in the residual loss and the distribution produced at inference answer different questions. Two evaluations of the exact denoiser can produce estimates with small residuals and a poor energy score. The learned O-S model, sampled with 50 evaluations, has a score close to D in this experiment. Thus, the approximation used in the residual term can misrepresent the conditional distribution. Its effect during training also depends on the data loss and sampling procedure.

Discreteness also matters for interval coverage. The exact conditional ensembles in Table 2 give coverage of 100.0% and 97.49% under the nominal 90% and 50% conventions, with mean widths 1.4581 and 1.3485 in standardized coordinates. The intervals include their endpoints, so tied values can raise coverage above the values expected for a continuous distribution. This explains why the nominal percentages alone would mischaracterize calibration in this toy.

## E Prediction scores and interval coverage

Here and in Appendix F, Gaussian denotes interpolation with a Gaussian residual fit on TRAIN. Gaussian regression instead uses the wider available context; Table 1 and Appendix G present that model separately.

The five primary quantities differ in both units and scale. Subtracting each quantity's TRAIN mean and dividing by its TRAIN scale makes field values in nT and scalar variance in $\mathrm{nT}^2$ dimensionless. The energy score then compares predictions for the whole missing hour in 60 coordinates, balancing distance to the withheld record against separation between predictions. The true conditional distribution minimizes the expected score [12]. Here, energy is the name of a statistical criterion. The magnetic energy density calculation below addresses a different physical question.

Each seed in Table 4 corresponds to one training run. All runs reach the common final update, while validation chooses the checkpoint used for prediction. It makes this choice separately for raw predictions and predictions after clipping. Coverage is calculated from each model's own ensemble before averaging across seeds.

The intervals in Table 5 quantify how score differences vary across the test years for the existing fits and samples. Each comparison uses the same resampled years for both methods. With seed 7003, 10,000 resamples take either the twenty individual years or the ten consecutive groups of two years, from 2005–2006 through 2023–2024. Repeated occurrences of a year each contribute their months, preserving equal weight per month.

For O-S minus D, the relative intervals are $[-0.016766, 0.014049]$ for individual years and $[-0.016842, 0.012995]$ for groups of two years. Both fit inside the original range of two percent in either direction. The intervals for O-S versus I-S and I-S versus D span both improvement and deterioration. These are marginal intervals for the existing fits and samples, based on twenty years or ten groups. Their interpretation across solar cycles depends on temporal stability, and their nominal coverage over repeated studies remains unverified. Variation from new training runs is outside this resampling calculation.

Table 4: Validation choice, primary energy score and raw nominal 90% coverage for each training seed. Validation selects checkpoints separately for raw and clipped predictions. Scores use standardized coordinates. Coverage is calculated from each ensemble of 39 samples, then averaged over five primary quantities and twelve hidden positions.

| | | Chosen updates | | Energy score | | Raw coverage |
|---|---|---|---|---|---|---|
| Method | Seed | Raw | Clipped | Raw | Clipped | (%) |
| D | 7 | 9560 | 29636 | 1.9575 | 1.9147 | 67.2 |
| D | 13 | 9560 | 9560 | 1.8297 | 1.8213 | 75.9 |
| D | 42 | 16252 | 16252 | 1.8758 | 1.8724 | 75.4 |
| I-S | 7 | 9560 | 9560 | 1.9423 | 1.9385 | 63.3 |
| I-S | 13 | 22944 | 22944 | 1.8954 | 1.8826 | 66.5 |
| I-S | 42 | 16252 | 16252 | 1.8070 | 1.8071 | 78.2 |
| O-S | 7 | 29636 | 29636 | 1.9558 | 1.9116 | 60.4 |
| O-S | 13 | 29636 | 29636 | 1.8934 | 1.8824 | 70.5 |
| O-S | 42 | 16252 | 16252 | 1.8056 | 1.8045 | 79.2 |
| G5 | 7 | 29636 | 29636 | 1.7236 | 1.7239 | 75.9 |
| G5 | 13 | 29636 | 29636 | 1.7853 | 1.7889 | 67.1 |
| G5 | 42 | 16252 | 16252 | 1.8453 | 1.8404 | 84.0 |

Table 5: Primary energy score differences, first method minus second. Unsuperscripted procedures use raw predictions; $c$ denotes clipping with a separate validation choice of checkpoint. The first four rows assess the original O-S superiority criterion; the others are secondary comparisons. Year and Two-year give marginal percentile 95% intervals using the same resampled years for both methods, with individual years or consecutive groups of two years. Seed differences appear in the order 7, 13, 42. Gaussian denotes interpolation with a Gaussian residual.

| Contrast | Mean | Year | Two-year | Seed differences |
|---|---|---|---|---|
| O-S − D | -0.0028 | $[-0.0346, 0.0242]$ | $[-0.0345, 0.0233]$ | $(-0.0018, 0.0637, -0.0702)$ |
| O-S − D$^c$ | 0.0154 | $[-0.0024, 0.0312]$ | $[-0.0034, 0.0320]$ | $(0.0410, 0.0721, -0.0669)$ |
| O-S − G5 | 0.1002 | $[0.0712, 0.1391]$ | $[0.0671, 0.1422]$ | $(0.2322, 0.1080, -0.0398)$ |
| O-S − G5$^c$ | 0.1005 | $[0.0716, 0.1396]$ | $[0.0677, 0.1420]$ | $(0.2319, 0.1045, -0.0348)$ |
| O-S − I-S | 0.0033 | $[-0.0236, 0.0349]$ | $[-0.0259, 0.0431]$ | $(0.0135, -0.0020, -0.0015)$ |
| I-S − D | -0.0061 | $[-0.0562, 0.0330]$ | $[-0.0720, 0.0410]$ | $(-0.0152, 0.0656, -0.0688)$ |
| O-S − Gaussian | -0.2402 | $[-0.5298, -0.0481]$ | $[-0.4539, -0.0882]$ | $(-0.1694, -0.2318, -0.3196)$ |

### E.1 Interval coverage, width, and score contributions

Similar primary scores conceal differences in interval coverage. O-S falls farther below 90% coverage than D by 2.8 percentage points, exceeding the prespecified threshold of two percentage points (Table 6). Gaussian interpolation is closer to 90%, but its 50% intervals contain substantially more than half of the measurements. These percentages describe coverage across the test set.

The differences also vary across quantities. O-S covers scalar variance more often than D, but covers each vector component and mean magnitude less often, as Table 7 shows. The energy score assesses their joint distribution across all 60 coordinates. Coordinate coverage instead locates errors in the individual quantities and positions.

G5's mean score improvement over D is largely accounted for by its larger spread term. Its mean distance to the withheld measurements is also slightly smaller. Figure 4(a) separates these contributions using the checkpoints that minimize raw validation score, the same 60 standardized coordinates, and equal month and seed weights.

The score can improve through different changes in these two terms. At seed 13, G5 reduces both distance to the measurements and spread; the larger decrease in distance lowers the score. At seed 42, both terms increase, and the larger increase in the spread term lowers the score. This arithmetic explains the score differences for the existing predictions. Representation and training loss change together in G5; varying them separately would distinguish their effects on learning.

**Where observations fall outside the intervals.** G5's nominal 90% intervals cover only 40.5% of the $v_s$ observations. To show the direction of the errors, the calculation separates observations below the second smallest of 39 samples, between the second and 38th inclusive, and above the 38th. When ensemble members and the observation are independent samples from the same continuous conditional distribution, each observation rank is equally likely. The expected fractions are then 5%, 90% and 5%.

For each fit, all TEST windows contribute: fractions are averaged over positions, then over windows in each month, with equal weights for months and for fits of the same method. Examining each variable and each tail complements the proper score [14]: an overall coverage percentage can combine errors in opposite directions.

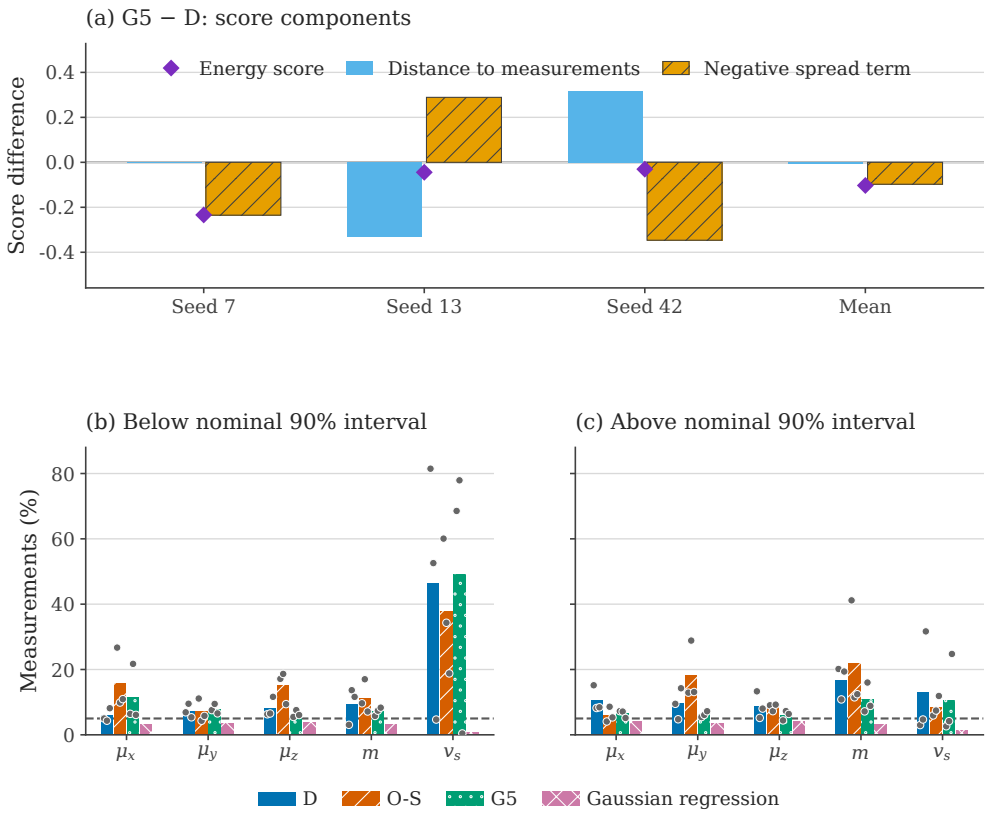
Across the three fits, 49.0% of observations fall below G5's $v_s$ intervals and 10.5% above them. Figure 4 shows why the mean alone is incomplete: errors are predominantly below the intervals at seeds 7 and 13, but above them at seed 42. D has the same reversal at seed 42. For O-S, the mean fractions are 37.7% below and 8.4% above.

For G5's $v_s$ intervals, the fraction below exceeds the fraction above by 38.4 percentage points. The 95% interval is $[36.2, 40.6]$, using 10,000 resamples of years and the same years for both fractions. Repeated occurrences of each year retain their weight. The difference uses percentages before rounding. This contrast quantifies temporal variation for the existing models, checkpoint choices and samples; new fits or new checkpoint choices would add other sources of uncertainty.

Gaussian regression covers 98.1% of $v_s$ observations, with 0.6% below and 1.2% above its intervals. Those wide intervals accompany a worse primary energy score. Reading coverage, width and score together distinguishes how often intervals contain the data, how concentrated the predictions are, and their performance under a proper scoring rule. Both temporal resampling schemes cover every method and coordinate.

Changing the coordinates in the score changes the question it asks. The vector score in Table 8 uses 36 standardized coordinates; adding $m$, $v_s$ and $v_v$ gives 72. Although $v_v$ follows from the other quantities for realizable statistics, including it changes distances and the relative influence of errors. The resulting scores have different scales and can rank methods differently. Gaussian interpolation has a lower mean vector score than D, I-S and O-S. G5 has the lowest mean primary score over five quantities, while its rank changes when $v_v$ enters the score.

The magnetic energy calculation assesses a different aspect of the prediction. For realizable statistics, $m^2 + v_s$ equals the mean squared magnitude of the five vector averages, each over one minute. Conversion to $\mathrm{T}^2$ and division by $2\mu_0$ give the magnetic energy density represented by those averages; variation within the original minute averages is absent. The same scalar transformation is applied to every predicted bin and averaged over the missing hour for each ensemble member. CRPS compares this scalar ensemble with the energy density of the withheld record. Interpreting each predicted value as the energy density of vectors with the other predicted moments depends on their joint realizability. CRPS and interval width have units of $\mathrm{J\,m^{-3}}$, shown as multiples of $10^{-12}$ in Table 8. Raw Gaussian interpolation has the lowest CRPS for this quantity among the procedures in that table. Clipping brings the moments into the ranges in Equation 3 to rounding tolerance, but its effect on the score depends on how the predictive distribution changes.




Bars: mean of three diffusion fits or one Gaussian. Gray dots: individual fits.

Figure 4: Why scores and interval coverage differ. Diffusion results use the checkpoints that minimize raw validation score. (a) G5 minus D for each seed and their mean. The change in distance to the withheld record, less the change in the spread term, gives the energy score difference. Both terms use 60 coordinates standardized on TRAIN; spread is half the mean distance between distinct predictions. (b,c) Fractions of observations below and above nominal 90% intervals for the five primary quantities. Diffusion bars average three fits; Gaussian regression uses one ensemble per window. Gray dots show seeds 7, 13 and 42 from left to right. The dashed 5% line assumes that, at each time and coordinate, the observation and ensemble members are exchangeable, with no ties. Months receive equal weight. This analysis uses the existing TEST ensembles after the original results were available.

### E.2 Vector increments at the edges and inside the missing interval

At the two edges of the missing hour, vector increments connect indices 17 to 18 and 29 to 30. Table 9 gives their mean magnitudes and the RMSE of individual predictions, calculated across 39 samples within each window and then averaged over months and seeds. This RMSE combines error of the predictive mean with predictive spread. It describes edge behavior, while the proper scores assess distributional accuracy.

The withheld record has mean increment magnitudes of 1.0460 nT at the left edge, 1.0539 nT at the right edge, and 1.0831 nT inside the missing hour. Interpolation imposes a smooth connection between the available measurements. Comparison with the withheld increments tests how well that smooth connection represents the actual variation.

The variogram score examines dependence through differences between coordinates, using exponent $1/2$ [15]. This calculation weights 175 pairs equally: 55 connect adjacent times within a primary quantity and 120 connect different quantities at the same time. For pair $(i, j)$, let $d_k = |x_{k,i} - x_{k,j}|^{1/2}$. The contribution is $(|y_i - y_j|^{1/2} - \bar{d})^2 - s_d^2/K$, with $K = 39$ and variance $s_d^2$ computed using

Table 6: Marginal continuous ranked probability scores, interval coverage and width for the five primary quantities. Scores and widths use dimensionless standardized values; coverage is in percent. Results average hidden positions, months and three diffusion seeds. Each other method uses one ensemble. Clipped rows apply the clipping map. Diffusion procedures choose checkpoints separately before and after clipping, except the D row that retains the weights from the raw procedure. Gaussian denotes interpolation with a Gaussian residual.

| Procedure | Continuous ranked probability score | 50% intervals | | 90% intervals | |
|---|---|---|---|---|---|
| | | Coverage (%) | Width | Coverage (%) | Width |
| D raw | 0.1753 | 35.5 | 0.2878 | 72.8 | 0.8709 |
| D clipped | 0.1721 | 33.5 | 0.2814 | 71.0 | 0.8617 |
| D clipped, raw weights | 0.1734 | 36.1 | 0.2883 | 73.5 | 0.8722 |
| I-S raw | 0.1745 | 32.7 | 0.2785 | 69.4 | 0.8315 |
| I-S clipped | 0.1735 | 32.8 | 0.2789 | 69.6 | 0.8319 |
| O-S raw | 0.1729 | 31.9 | 0.2691 | 70.0 | 0.8213 |
| O-S clipped | 0.1701 | 32.6 | 0.2685 | 71.1 | 0.8195 |
| G5 raw | 0.1634 | 38.1 | 0.3137 | 75.7 | 0.9308 |
| G5 clipped | 0.1624 | 38.4 | 0.3128 | 75.8 | 0.9287 |
| Gaussian raw | 0.1960 | 76.4 | 0.6262 | 94.3 | 1.5822 |
| Gaussian clipped | 0.1935 | 68.6 | 0.5512 | 93.6 | 1.3933 |
| Interpolation raw | 0.1993 | 0.0 | 0.0000 | 0.0 | 0.0000 |
| Interpolation clipped | 0.1993 | 0.0 | 0.0000 | 0.0 | 0.0000 |
| Persistence raw | 0.2697 | 0.1 | 0.0000 | 0.1 | 0.0000 |
| Persistence clipped | 0.2697 | 0.1 | 0.0000 | 0.1 | 0.0000 |

Table 7: Nominal 90% coverage in percent for each primary quantity before clipping. Hidden positions, months and diffusion seeds receive equal weight in their respective averages. Gaussian denotes interpolation with a Gaussian residual.

| Procedure | $\mu_x$ | $\mu_y$ | $\mu_z$ | $m$ | $v_s$ |
|---|---|---|---|---|---|
| D | 83.6 | 83.2 | 83.0 | 73.8 | 40.6 |
| I-S | 82.2 | 74.8 | 80.7 | 74.3 | 34.9 |
| O-S | 78.2 | 74.6 | 76.4 | 67.0 | 53.9 |
| G5 | 82.1 | 85.9 | 87.6 | 82.2 | 40.5 |
| Gaussian | 93.1 | 93.5 | 92.2 | 94.1 | 98.5 |
| Interpolation | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Persistence | 0.1 | 0.2 | 0.1 | 0.0 | 0.0 |

divisor $K - 1$. The squared term compares the transformed difference in the measurements with the ensemble mean. For independent samples, that estimate has variance $\mathrm{Var}(d_k)/K$; subtracting $s_d^2/K$ removes its upward Monte Carlo bias in expectation. The final score averages these contributions. It complements the energy score by emphasizing differences between coordinates.

### E.3 Numerical precision when transforming observations

Normalization and its inverse introduce rounding error even in physically realizable data. Table 10 compares these errors in the 5,760 hidden observation bins. The affine transform uses TRAIN statistics. One calculation retains double precision throughout; another converts the standardized values to single precision before returning to double precision for the inverse transform and scoring. The representation with five coordinates then obtains $v_v$ by algebraic completion. Every route preserves the inequalities for these observations. Computing $v_v$ this way can make the equality residual zero while rounding error remains in the primary coordinates.

Table 8: Scores for alternative magnetic quantities. The vector and six-coordinate energy scores use 36 and 72 standardized coordinates. The last three columns assess the mean magnetic energy density over the missing hour, computed for each ensemble member from $m^2 + v_s$. This quantity retains the energy represented by vector averages over one minute and excludes variation within those averages. Raw rows evaluate the same scalar even when the other predicted moments are inconsistent. Continuous ranked probability score and interval width are in $10^{-12}$ J m$^{-3}$; coverage is in percent. Validation chooses model states separately for raw and clipped diffusion predictions. Gaussian denotes Gaussian interpolation.

| | Energy score | | Resolved magnetic energy density | | |
|---|---|---|---|---|---|
| Procedure | Vector | Six coordinates | Continuous ranked probability score | 90% coverage (%) | 90% width |
| D raw | 1.6351 | 2.1528 | 2.8599 | 74.3 | 6.2166 |
| D clipped | 1.6222 | 2.5726 | 2.7568 | 68.5 | 6.6024 |
| I-S raw | 1.6435 | 2.1590 | 2.4664 | 72.4 | 6.6657 |
| I-S clipped | 1.6435 | 2.8791 | 2.3049 | 73.1 | 6.8576 |
| O-S raw | 1.6316 | 2.1503 | 2.9225 | 65.3 | 6.5977 |
| O-S clipped | 1.6316 | 2.3863 | 2.7461 | 72.2 | 6.6863 |
| G5 raw | 1.5740 | 2.4959 | 1.8961 | 82.2 | 6.8060 |
| G5 clipped | 1.5740 | 2.4285 | 1.9066 | 82.4 | 6.8757 |
| Gaussian raw | 1.5909 | 2.7342 | 1.0816 | 93.8 | 6.1012 |
| Gaussian clipped | 1.5909 | 4.2146 | 2.5465 | 64.8 | 11.2941 |
| Interpolation raw | 1.9834 | 2.8920 | 1.2771 | 0.0 | 0.0000 |
| Interpolation clipped | 1.9834 | 4.6381 | 1.2771 | 0.0 | 0.0000 |
| Persistence raw | 2.7925 | 3.6601 | 2.6284 | 0.0 | 0.0000 |
| Persistence clipped | 2.7925 | 3.6601 | 2.6284 | 0.0 | 0.0000 |

Table 9: Vector increments and variogram scores before clipping. Increment magnitudes and RMSE use nT. Observed mean increments are 1.0460 at the left edge, 1.0539 at the right edge and 1.0831 inside the missing hour. RMSE is calculated across the 39 individual predictions in each window, then averaged over months and seeds; it includes both predictive spread and error of the predictive mean. The variogram uses the 175 standardized coordinate pairs defined in the text and the finite-ensemble bias correction. Clipping retains the vector increments when the model weights stay the same, while its changes to $m$ and $v_s$ can alter the variogram. D also changes vector predictions through the separate validation choice of checkpoint at seed 7. Gaussian denotes interpolation with a Gaussian residual.

| | | Mean increment magnitude | | | Root mean square error | |
|---|---|---|---|---|---|---|
| Procedure | Variogram | Left edge | Right edge | Internal | Left edge | Right edge |
| D | 0.0524 | 1.3121 | 1.3581 | 0.8560 | 1.8164 | 1.8323 |
| I-S | 0.0521 | 1.3309 | 1.4224 | 0.8156 | 1.8059 | 1.8854 |
| O-S | 0.0524 | 1.3111 | 1.3296 | 0.7635 | 1.7974 | 1.7990 |
| G5 | 0.0490 | 1.3717 | 1.3830 | 1.0789 | 1.8893 | 1.8958 |
| Gaussian | 0.0905 | 1.5737 | 1.5587 | 1.5718 | 2.1020 | 2.1114 |
| Interpolation | 0.0658 | 0.2518 | 0.2518 | 0.2518 | 1.0373 | 1.0309 |
| Persistence | 0.1003 | 0.0000 | 3.2730 | 0.0000 | 1.0460 | 3.1140 |

# F Variation across spacecraft, years, and magnetic conditions

Table 11 examines how the primary score varies with spacecraft, calendar period and magnetic conditions. For each subgroup, the score averages first over windows within a month and then equally over the months present. Combining subgroup scores by window count can give a different result from the overall test score.

Spacecraft and calendar period vary together in this sample. In 2020–2024, 119 windows use Wind and only one uses ACE, so the spacecraft comparison also includes temporal variation. The calendar groups themselves span different phases of the solar cycle. The intervals from resampling years

Table 10: Numerical error after normalization and its inverse for the 5,760 hidden observation bins. Single precision denotes storage of normalized values; normalization and its inverse use double precision. Maximum $|\Delta u|$ is the largest absolute change among the five standardized primary coordinates in all hidden bins. Equality residuals are in nT$^2$. Deriving $v_v$ makes the computed residual zero without removing rounding error from the primary coordinates.

| Representation | Maximum $\lvert\Delta u\rvert$ | Mean $\lvert R_{\mathrm{obs}}\rvert$ | Maximum $\lvert R_{\mathrm{obs}}\rvert$ |
|---|---|---|---|
| Original, double precision | 0 | $8.476 \times 10^{-15}$ | $3.326 \times 10^{-13}$ |
| Double precision round trip | $1.271 \times 10^{-15}$ | $8.987 \times 10^{-15}$ | $5.573 \times 10^{-13}$ |
| Single precision, six coordinates | $4.977 \times 10^{-7}$ | $1.571 \times 10^{-6}$ | $6.268 \times 10^{-5}$ |
| Single precision, five coordinates | $4.977 \times 10^{-7}$ | 0 | 0 |

describe the existing predictions and remain exploratory, without adjustment for the number of comparisons.

Two further comparisons group windows by conditions during the missing hour. The 480 TEST windows are divided into thirds by mean withheld magnitude, using thresholds of 3.9350 and 5.7891 nT, or by mean internal vector increment, using thresholds of 0.6134 and 1.2055 nT. Each division gives 160 windows per group; none of the values equals a threshold.

Selecting cases by the withheld outcome changes the distribution being assessed. The forecast describes possible outcomes given the surrounding measurements; a subset chosen after seeing the missing values represents only some of those outcomes. Thus, its ranking under a proper score can change even when the forecast was correct for the original conditioning information. The opposite signs of O-S minus D at low and high magnitude describe these TEST subsets. Interpreting the difference as skill conditional on field strength involves accounting for that selection. Table 12 instead lists individual years; resampling intervals above concern groups spanning several years.

Table 11: Raw primary energy score across spacecraft, calendar periods, and thirds of TEST defined by withheld mean magnitude or mean internal vector increment. The last column gives a marginal 95% interval for O-S minus D from resampling years. Scores use standardized primary coordinates and average the same three diffusion fits. Groups defined by withheld values compare scores after selection on the outcome. Predictive superiority conditional on the available measurements concerns the outcome distribution before that selection. ACE denotes the Advanced Composition Explorer.

| Stratum | Windows | D | O-S | G5 | O-S − D | Year interval |
|---|---|---|---|---|---|---|
| Wind (52) | 335 | 1.7136 | 1.7119 | 1.6159 | -0.0017 | $[-0.0382, 0.0303]$ |
| ACE (71) | 145 | 1.8597 | 1.9008 | 1.7886 | 0.0411 | $[-0.0009, 0.0869]$ |
| 2005–2009 | 120 | 1.9817 | 2.0219 | 1.9051 | 0.0403 | $[0.0158, 0.0638]$ |
| 2010–2015 | 144 | 1.7735 | 1.7728 | 1.6890 | -0.0007 | $[-0.0416, 0.0415]$ |
| 2016–2019 | 96 | 1.6349 | 1.6220 | 1.5657 | -0.0129 | $[-0.0543, 0.0237]$ |
| 2020–2024 | 120 | 2.1330 | 2.0928 | 1.9545 | -0.0403 | $[-0.1384, 0.0293]$ |
| Magnitude, lower third | 160 | 1.1777 | 1.2324 | 1.1010 | 0.0547 | $[0.0195, 0.0835]$ |
| Magnitude, middle third | 160 | 1.4606 | 1.4747 | 1.4339 | 0.0140 | $[-0.0044, 0.0310]$ |
| Magnitude, upper third | 160 | 3.3910 | 3.3366 | 3.1849 | -0.0544 | $[-0.1089, -0.0023]$ |
| Increment, lower third | 160 | 0.8924 | 0.9174 | 0.8255 | 0.0250 | $[-0.0056, 0.0560]$ |
| Increment, middle third | 160 | 1.4411 | 1.4416 | 1.3768 | 0.0005 | $[-0.0392, 0.0434]$ |
| Increment, upper third | 160 | 3.2522 | 3.2284 | 3.0924 | -0.0238 | $[-0.0804, 0.0271]$ |

### F.1 Directional cancellation with thresholds from TRAIN

Directional cancellation occurs when the norm of the mean vector is smaller than the mean of the individual magnitudes. The cancellation analysis examines whether the variance identity penalty behaves differently under these conditions. For each bin, define $C = (m^2 - \|\mu\|^2)/m^2$, with $C = 0$ when $m = 0$, and $A = (m^2 - \|\mu\|^2)/B_{\mathrm{ref}}^2$. Both measures are dimensionless: $C$ expresses cancellation relative to the bin's squared mean magnitude, while $A$ uses the common TRAIN scale.

Table 12: Annual raw primary energy scores in standardized coordinates. Each row uses 24 test windows, averages the year's two TEST months and then the three diffusion seeds. Gaussian denotes the shared ensemble from Gaussian interpolation.

| Year | D | I-S | O-S | G5 | Gaussian |
|---|---|---|---|---|---|
| 2005 | 3.3402 | 3.4233 | 3.3349 | 3.1962 | 3.3435 |
| 2006 | 1.8130 | 1.8449 | 1.8911 | 1.8222 | 2.0005 |
| 2007 | 1.5185 | 1.6045 | 1.5749 | 1.4551 | 2.1359 |
| 2008 | 1.8584 | 1.8789 | 1.8882 | 1.7878 | 2.1347 |
| 2009 | 1.3782 | 1.4624 | 1.4206 | 1.2640 | 1.5893 |
| 2010 | 1.2774 | 1.3131 | 1.2785 | 1.2081 | 1.5417 |
| 2011 | 2.5543 | 2.5698 | 2.5144 | 2.3807 | 2.2021 |
| 2012 | 1.3628 | 1.3747 | 1.4155 | 1.3091 | 1.6150 |
| 2013 | 1.1809 | 1.2399 | 1.2585 | 1.1578 | 1.4643 |
| 2014 | 2.5162 | 2.4384 | 2.4978 | 2.4404 | 2.4855 |
| 2015 | 1.7493 | 1.7088 | 1.6719 | 1.6383 | 1.7181 |
| 2016 | 1.9325 | 1.8200 | 1.8618 | 1.7919 | 1.9331 |
| 2017 | 1.7522 | 1.7971 | 1.7414 | 1.7083 | 2.0006 |
| 2018 | 1.5171 | 1.5611 | 1.5522 | 1.4413 | 1.6696 |
| 2019 | 1.3378 | 1.3602 | 1.3326 | 1.3214 | 1.6536 |
| 2020 | 1.5016 | 1.5137 | 1.5409 | 1.4375 | 1.6264 |
| 2021 | 1.3806 | 1.4460 | 1.4136 | 1.3686 | 1.5898 |
| 2022 | 1.9837 | 2.0144 | 1.9358 | 1.9206 | 1.9962 |
| 2023 | 2.5198 | 2.3208 | 2.5221 | 2.1314 | 2.1605 |
| 2024 | 3.2793 | 2.9392 | 3.0514 | 2.9142 | 5.6425 |

The groups use mean $C$ over the 36 available bins or mean $A$ over the twelve hidden bins. High cancellation means strictly above the TRAIN 75th percentile: 0.044395 for the available intervals and 0.022696 for the missing interval. Every one of the 122,309 overlapping training windows contributes equally to these thresholds, whereas normalization uses distinct bins. The rules predate the implementation, which followed inspection of the primary results; the analysis is exploratory.

Window means of $Q/B_{\mathrm{ref}}^2$ define the energy groups. Here $Q = m^2 + v_s$ is the magnetic second moment in $\mathrm{nT}^2$, and the ratio is dimensionless; this physical quantity is distinct from the statistical energy score. TRAIN quartile thresholds are $(0.349809, 0.571606, 1.078729)$ for the available intervals and $(0.343518, 0.563106, 1.069013)$ for the missing interval. A value equal to an energy threshold enters the higher energy group. A value equal to the cancellation threshold enters the rest group.

The 21 groups comprise the full test set, high cancellation and the rest, and combinations of cancellation with energy conditions. All 18 procedures are evaluated in each group. Scores give equal weight to the months present and then average the three diffusion seeds.

High cancellation in the available measurements identifies 128 windows across all twenty years. O-S minus D is $-0.0256$, with a 95% interval of $[-0.0740, 0.0147]$ from resampling years. All three seed differences are negative, while the interval includes zero. O-S minus G5 is $0.0573$, with interval $[0.0053, 0.1246]$, and O-S minus D is $0.0114$ in the remaining 352 windows. G5 has a lower mean score in these 128 windows, while the O-S versus D interval spans zero. The group uses information available to the model; cancellation can still vary with other magnetic conditions.

High cancellation during the missing hour identifies 121 windows. O-S minus D is $-0.0128$, with interval $[-0.0858, 0.0629]$, compared with $0.0068$ in the remaining 359 windows. Although the thresholds come from TRAIN, assignment to these TEST groups uses the withheld values. The score comparison is consequently conditional on outcome selection.

Combining high cancellation during the missing hour with the lowest energy quartile leaves seven windows from five years. Their scores describe this small subset of TEST outcomes. Interpreting them as evidence of conditional predictive superiority depends on accounting for the rule that uses withheld values to form the groups.

# G Clipping samples to satisfy moment constraints and Gaussian regression

## G.1 Algebraic completion and clipping of existing samples

Algebraic completion changes only $v_v$. Because the primary energy score uses the other five quantities, it has exactly the same value for every ensemble before and after completion. The following comparisons retain the models, samples and windows, using model weights chosen by raw validation score. They isolate the effect of changing existing predictions. Training G5 also changes the representation and loss, so its score comparison includes those effects.

Clipping changes the primary quantities when they fall outside the realizable ranges. It retains $\mu$, raises $m$ to $r = \|\mu\|$ when necessary, and clips $v_s$ to $[0, 1.5m^2 + 2.5r^2]$ using the new value of $m$. Algebraic completion then gives $v_v$. The resulting moments are realizable, and repeating the operation leaves them the same to numerical tolerance. This sequential construction can move samples farther than a nearest-point projection.

The change in score is limited by how far clipping moves the samples. For primary samples $X = (x_1, \ldots, x_K)$ and observation $y$,

$$|\mathrm{ES}(CX, y) - \mathrm{ES}(X, y)| \leq \frac{2}{K} \sum_{i=1}^{K} \|Cx_i - x_i\|. \tag{8}$$

By the reverse triangle inequality, each score term changes in absolute value by at most the mean sample displacement. Adding these contributions gives Equation 8; the argument applies to convex and nonconvex realizable sets. The bound concerns magnitude, so the score can increase or decrease. Distances use all 60 primary coordinates after TRAIN standardization and are dimensionless. Table 13 adds the change in distance to withheld measurements and subtracts the change in the spread term.

Table 13: Effects of clipping the same ensembles before any further model selection. Each comparison applies clipping to an existing ensemble, with the same model weights and evaluation windows; diffusion uses the models chosen by raw validation score. Months receive equal weight, followed by an average over three diffusion seeds or the one comparison ensemble. Distances use 60 primary coordinates standardized on TRAIN. Displacement is the mean distance moved by an ensemble member. The two term changes, with signs as shown, sum to the change in energy score before rounding.

| Procedure | Displacement | $\Delta$ distance | $-\Delta$ spread term | $\Delta$ energy score |
|---|---|---|---|---|
| D | 0.1254 | -0.00350 | -0.00287 | -0.00637 |
| I-S | 0.0997 | -0.00427 | -0.00119 | -0.00547 |
| O-S | 0.1521 | -0.01952 | 0.00080 | -0.01873 |
| G5 | 0.1055 | 0.00003 | -0.00038 | -0.00035 |
| Gaussian interpolation | 2.1403 | -0.32056 | 0.42478 | 0.10422 |

The reported failure rates count departures beyond rounding tolerance without distinguishing their size. Before any allowance for rounding, $\max(r - m, 0)/B_{\mathrm{ref}}$ measures the dimensionless shortfall in mean magnitude. Its 90th percentile within each window averages to 0.01648 for G5 and 0.34407 for Gaussian interpolation. For negative scalar variance, the corresponding quantity is $\max(-v_s, 0)/B_{\mathrm{ref}}^2$, with averages of 0.000342 and 0.017372. A value of zero means that the respective inequality holds.

For each window, the percentile is calculated across its samples and hidden bins. These window percentiles are averaged within each month, then across months and seeds. Taking one percentile across all windows would be a different calculation. G5 and Gaussian interpolation both frequently fail the inequalities, yet clipping moves their samples by very different distances. G5's small score change coexists with many samples outside the realizable set.

For D, Table 13 attributes an energy score change of $-0.00637$ to clipping the ensemble chosen by raw validation score. Choosing the model independently on VAL after clipping adds a further change of $-0.01183$; seed 7 is the only run for which the chosen training step changes. The raw and clipped scores in Table 1 combine these two effects: altering the samples and choosing the model.

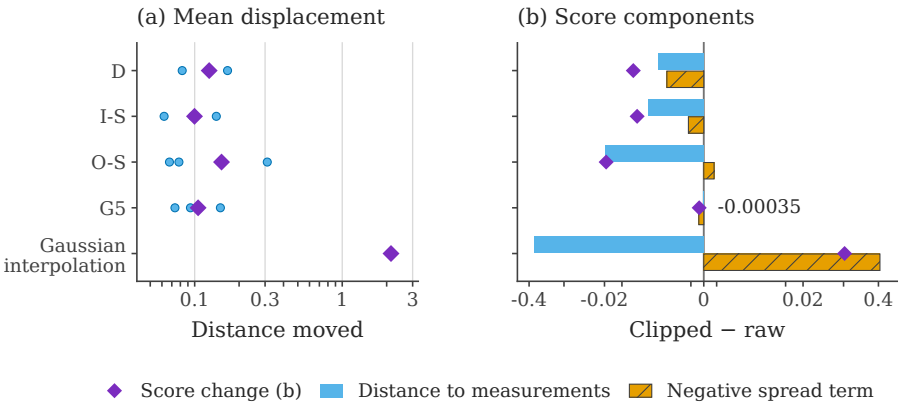



Figure 5: Clipping samples from models chosen by raw validation score. (a) Mean displacement in 60 standardized coordinates on a logarithmic axis. Blue circles show individual fits; purple diamonds show their mean or one Gaussian ensemble. (b) Solid bars show the change in distance to withheld values; striped bars show the change in the negative spread term. Their sum, the energy score change, appears as purple diamonds. The symmetric logarithmic axis is linear inside $\pm 0.005$. Clipping changes D's score by $-0.0064$; choosing another checkpoint adds $-0.0118$.

### G.2 Multivariate Gaussian regression

Gaussian regression uses the measurements before and after the missing hour to estimate departures from interpolation. Its input $X$ contains all 216 standardized available coordinates; $Y$ contains the 60 primary residuals from interpolation between bins 17 and 30, divided by the corresponding TRAIN scales. Fitting $W$ and an unpenalized intercept $b$ on TRAIN minimizes $\|Y - XW - b\|_F^2/N + \lambda\|W\|_F^2$, with equal weight for all 122,309 overlapping windows.

The nine parameter combinations use $\lambda \in \{0.001, 0.1, 1\}$ and covariance $(1-\rho)S + \rho\,\mathrm{diag}(S) + 10^{-6}I$, where $\rho \in \{0, 0.5, 1\}$ and $S$ is the empirical residual covariance with divisor $N$. Within each fit, the conditional mean varies with context and the residual covariance is common to all contexts. The 60 primary coordinates have a joint Gaussian distribution; deriving $v_v$ gives a non-Gaussian distribution over all six quantities. Positive definiteness makes the Gaussian covariance valid, while realizability remains a separate condition on its samples.

Common random innovations allow the nine parameter combinations to be compared using the same noise. Each VAL window uses 39 innovations from PCG64, identified by split, window, sample and seed 13. Validation minimizes energy score with equal month weights, separately before and after clipping; exact ties follow lexicographic $(\lambda, \rho)$ order. Both procedures choose $\lambda = 0.1$ and $\rho = 0$, with raw and clipped VAL scores of 1.73275 and 1.83453, respectively. These choices precede generation of the single TEST ensemble.

Gaussian regression changes three parts of the interpolation baseline: the available measurements used as input, the fit of the conditional mean, and the finite realization of the sampling noise. Its score difference measures their combined effect.

G5 has a lower mean energy score than Gaussian regression: the difference is $-0.1503$, with a year interval of $[-0.2302, -0.0829]$ and an interval of $[-0.2105, -0.0913]$ from resampling groups of two years. Both intervals lie below zero. The corresponding O-S difference is $-0.0502$, with intervals $[-0.1336, 0.0327]$ and $[-0.0913, -0.0080]$, respectively. For O-S, the interval spans zero when resampling individual years and excludes it when resampling groups of two years. Changing the temporal grouping changes the resampling distribution and the uncertainty in this comparison.

The marginal 95% intervals use the original 10,000 temporal resamples, with the same years for both methods in each comparison. They measure variation across the test years while retaining the three diffusion fits, the Gaussian ensemble and the validation choices. Repeating training, noise generation or validation choice would introduce further variation. This analysis uses a test period examined during earlier development.

Before clipping, Gaussian regression has a vector energy score of 1.5728 over 36 standardized coordinates and a CRPS of $1.0497 \times 10^{-12}$ J m$^{-3}$ for mean resolved magnetic energy density. These scores concern different predictions and use different scales. Clipping leaves the vector predictions intact but raises the energy-density CRPS to $2.1873 \times 10^{-12}$ J m$^{-3}$. The mean displacement over all 60 standardized primary coordinates is 1.7733.

For both Gaussian baselines, clipping brings samples closer to the withheld measurements but reduces the spread term by more. Subtracting that smaller term raises the primary energy score. The samples now satisfy the moment constraints, yet their distribution receives a worse score. Interval coverage answers how often individual intervals contain measurements; the energy score also responds to the joint arrangement of the predictions.

## H Penalty strength after training

The coefficient comparison evaluates every model after 29,636 training steps, using $c_R \in \{0, 0.00005, 0.0005, 0.005\}$ at seeds 7, 13 and 42 in Table 14. Zero gives D, and positive coefficients apply O-S's variance identity penalty. The final D and O-S models contribute six fits in total; the two additional coefficients contribute six more. Evaluation uses the same 480 windows, 39 samples and 50 DDIM evaluations as the main experiment. Each comparison evaluates an ensemble before and after clipping.

The final training step was specified before this comparison, and all twelve fits are included. These TEST scores serve only to compare the coefficient settings at that stage. Equal update counts can involve different amounts of computation, and score differences can reflect convergence or overfitting. This makes the experiment a comparison at a common training stage.

Table 14: Penalty coefficients compared after 29,636 training steps. Each row describes one fit and gives equal weight to the months present. Energy scores use the five primary quantities at twelve times in standardized coordinates. For samples before clipping, the remaining columns give mean absolute $R_{\mathrm{obs}}$ in nT$^2$, the percentage of bins failing a moment inequality beyond rounding tolerance, nominal 90% coverage and mean interval width in standardized units. The inequality percentage excludes the equality test. All twelve coefficient/seed combinations are shown at the common final update, without selecting a coefficient or checkpoint from these TEST scores.

| | Energy score | | Raw samples | | | |
|---|---|---|---|---|---|---|
| Coefficient / seed | Raw | Clipped | $\lvert R \rvert$ | Bins failing inequalities (%) | Coverage (%) | Width |
| 0 / 7 | 1.9787 | 1.9147 | 3.5152 | 78.4 | 57.5 | 0.8018 |
| 0 / 13 | 1.9778 | 1.9515 | 3.1092 | 68.1 | 67.9 | 0.8369 |
| 0 / 42 | 1.9900 | 1.9847 | 4.2064 | 11.4 | 60.6 | 0.7955 |
| 0.00005 / 7 | 2.0412 | 1.9714 | 3.5544 | 80.4 | 57.4 | 0.7830 |
| 0.00005 / 13 | 1.8735 | 1.8653 | 2.8988 | 59.7 | 69.4 | 0.8347 |
| 0.00005 / 42 | 1.9757 | 1.9702 | 3.8648 | 15.5 | 62.9 | 0.8396 |
| 0.0005 / 7 | 1.9558 | 1.9116 | 2.8245 | 88.8 | 60.4 | 0.7797 |
| 0.0005 / 13 | 1.8934 | 1.8824 | 2.3812 | 32.5 | 70.5 | 0.8112 |
| 0.0005 / 42 | 1.9312 | 1.9284 | 3.1843 | 15.6 | 66.8 | 0.8634 |
| 0.005 / 7 | 2.0224 | 1.9934 | 1.5068 | 69.1 | 57.2 | 0.7742 |
| 0.005 / 13 | 2.0201 | 2.0114 | 1.7024 | 54.4 | 61.4 | 0.8303 |
| 0.005 / 42 | 1.8119 | 1.8036 | 1.1243 | 44.2 | 68.6 | 0.7964 |

A stronger penalty reduces the mean absolute residual, but its effect on predictive accuracy changes with its strength (Figure 6). The mean energy score first falls and then rises. The strongest penalty gives the smallest residual; the original coefficient gives the lowest mean score. At $c_R = 0.005$, the score is lower than D's at seed 42 and higher at seeds 7 and 13. Across training runs, reducing the residual accompanies different changes in the score.

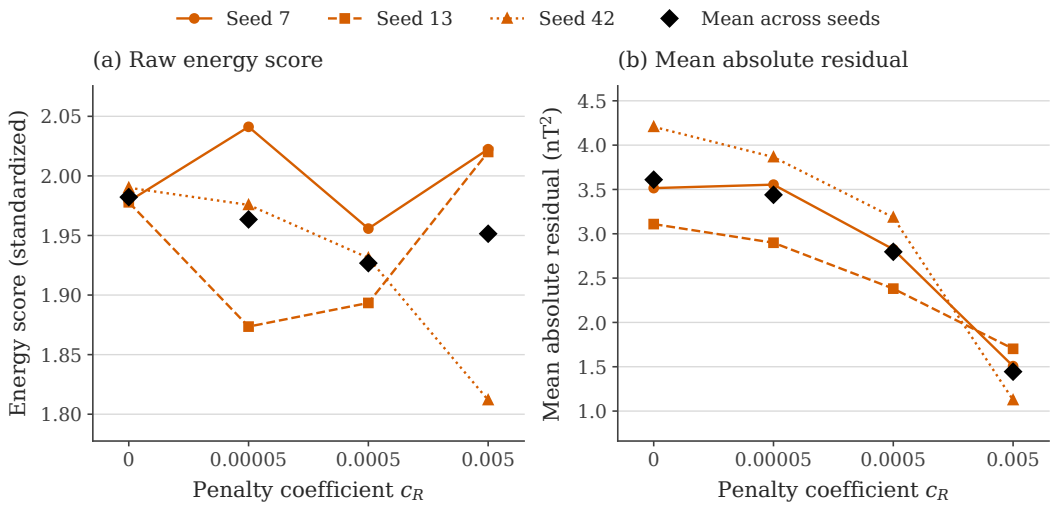



Figure 6: Energy score and residual of the variance identity after 29,636 training steps for the twelve models in Table 14. Lines connect each seed's four coefficient settings; solid black diamonds show the mean across three seeds. The horizontal axis treats the coefficient settings as equally spaced categories. The left subplot gives raw energy score in standardized primary coordinates, and the right gives mean absolute $R_{\text{obs}}$ in $\text{nT}^2$. Each model value gives equal weight to the months present.

At the original coefficient, O-S has a lower energy score than D at the end of all three training runs. Validation chooses earlier models in some runs, and the resulting O-S minus D differences change sign across seeds. The comparison consequently depends on which stage of training supplies the predictions. These three runs establish that dependence in the present experiment.

# I Related work on physical constraints in prediction and generation

Solar wind reconstruction addresses quantities at several scales. Gaussian process regression estimates upstream plasma and magnetic conditions at Mars between MAVEN observations, with uncertainty that depends on the available sampling [16]. Feedforward networks estimate second-order structure functions from incomplete magnetic records [1], while conditional diffusion generates future coronal EUV images [17]. Here, prediction concerns magnetic moments across a missing hour, conditional on measurements on both sides. Different orderings of the minute vectors give the same moments within each bin.

PISCES penalizes the difference between squared magnetic magnitude and squared component norm in an autoencoder for solar wind anomaly detection [18]. Its relation applies to reconstructions at each minute. Here, mean magnitude and mean vector are distinct statistics over bins of five minutes, linked by the variance identity in Equation 2.

Bastek et al. [3, Section 3.3 and Appendix A.6.1] distinguish a residual at the denoiser mean from one at a sample, and demonstrate collapse when training uses the residual alone. The HMM here makes the denoiser available by calculation, allowing direct examination of the sampling approximation. The solar wind experiments examine the corresponding training objective at several stages. Completing or clipping the same samples separates algebraic operations from changes during training.

Beucler et al. [5] compare losses over all outputs, including those from conservation layers, with training on direct outputs followed by algebraic completion. G5 uses the latter construction in a conditional generator, changing both representation and loss. Completing $v_v$ alone preserves the primary score, which uses the other five quantities. In climate downscaling, exact aggregation and nonnegativity impose distinct constraints, with separate assessments of predictive accuracy [19].

Nonlinear averaging also enters physical observation models. PGDM compares arithmetic temperature averaging with the fourth root of the mean fourth power, neglecting emissivity variation in the latter rule [20]. It conditions on geophysical variables, trains with MSE, and evaluates physical

consistency afterward. WIND uses diffusion forcing for pretraining and moment matching for posterior guidance to reconstruct weather trajectories from observations that include spatial and temporal averages [21]. Guidance approximates the clean state conditional on the noisy state by a Gaussian and propagates its covariance through a local linearization of the observation operator. Both methods incorporate averaging in the observation model.

Residual weights can also adapt during training. PILD uses a Laplace likelihood whose scale depends on residual magnitude, estimating conditional moments of that magnitude from minibatch statistics with exponential averaging [22]. Physics-Based Flow Matching evaluates residuals after integrating the flow to $t = 1$ and combines gradients of the flow matching and residual losses through ConFIG [4]. These objectives offer alternatives to the constant residual coefficient examined here.

## J Two examples of magnetic reconstruction

Figures 7 and 8 show two missing hours from TEST in physical units. Windows 120 and 360, counting from zero, are the midpoints of the two halves of the ordered list of 480 windows. Position determines their choice independently of model error, although TEST had already been examined. Diffusion uses seed 7. Every method uses the fit that minimizes raw validation score; Gaussian regression samples use seed 13.

Every method receives the same surrounding measurements and is compared with the same withheld values. The colored curve joins the marginal medians of 39 samples at each time and coordinate. After sorting those samples in increasing order, values 2 and 38 form the inclusive nominal 90% interval. The joined medians can be irregular and can differ from every sampled trajectory. These curves and intervals concern each time and quantity separately; simultaneous coverage of an entire trajectory is a separate property. Thin lines show the first three sampled trajectories, ensemble members 0, 1 and 2.

The horizontal axes mark the starts of bins of five minutes, relative to the missing hour. Figures 7 and 8 show predictions before clipping. Within each figure, the methods share a vertical scale for each quantity. For $v_s$, a symmetric logarithmic scale accommodates both small measured variances and Gaussian predictions outside the physical range. The reconstructed statistics describe five minutes of measurements; different sequences of minute averages can produce the same statistics.

Available measurements retain their values during sampling. The plots show them separately from the twelve predicted bins, with short line breaks from $-5$ to $0$ minutes and from $55$ to $60$ minutes. Each neighboring bin contains a different set of five vectors. The variance identity constrains the moments inside a bin, while the conditional model learns their dependence across time. The elevated variance predictions are model errors; Figure 9 shows them on linear scales.

The scalar variance $v_s$ is the variance of the magnitudes of five vector averages, each taken over one minute. In both examples, diffusion generally places its median above the withheld variance, which often falls outside the nominal intervals. In TEST window 120, every diffusion method covers $v_s$ in zero of the twelve hidden bins. In window 360, D, O-S and G5 cover zero, one and two bins, respectively. The aggregate coverage results show the same shortfall from nominal coverage for this quantity.

Gaussian regression's wider intervals contain all twelve withheld values in both windows. The model uses one residual covariance for every context. Its Gaussian distribution also assigns probability to negative $v_s$, even though variance is nonnegative. These examples show both the broad uncertainty intervals and the impossible values that accompany them.

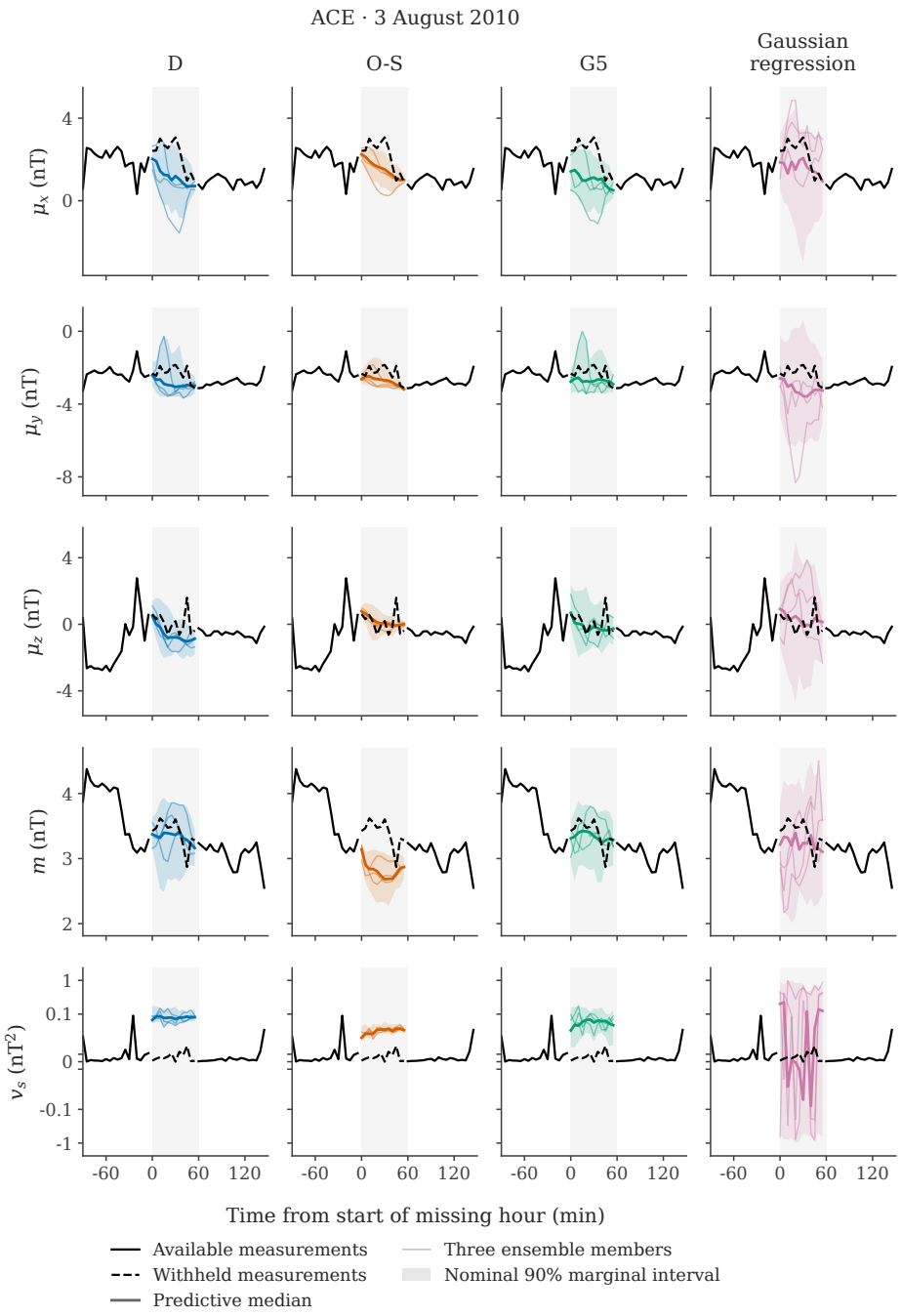



Figure 7: Reconstruction from ACE measurements on 3 August 2010, TEST window 120. The missing hour runs from 13:30 to 14:30 UTC on the OMNI arrival-time grid. Columns compare D, O-S, G5 and Gaussian regression, with model weights chosen by validation; rows show the five primary magnetic statistics in physical units. All samples appear before clipping. Diffusion uses seed 7 and Gaussian regression uses seed 13. Solid black lines show 90 minutes of available observations on each side of the shaded missing hour, and dashed black lines show the withheld measurements. Short line breaks separate available observations from predictions. Colored lines show marginal medians, bands show nominal 90% intervals, and thin lines show ensemble members 0, 1 and 2. Times mark the starts of bins of five minutes relative to the missing hour. The $v_s$ axis is linear within $0.05$ nT$^2$ of zero and logarithmic outside, so equal vertical distances can represent different absolute errors in that row. Appendix J explains the case choice and the construction of the intervals.

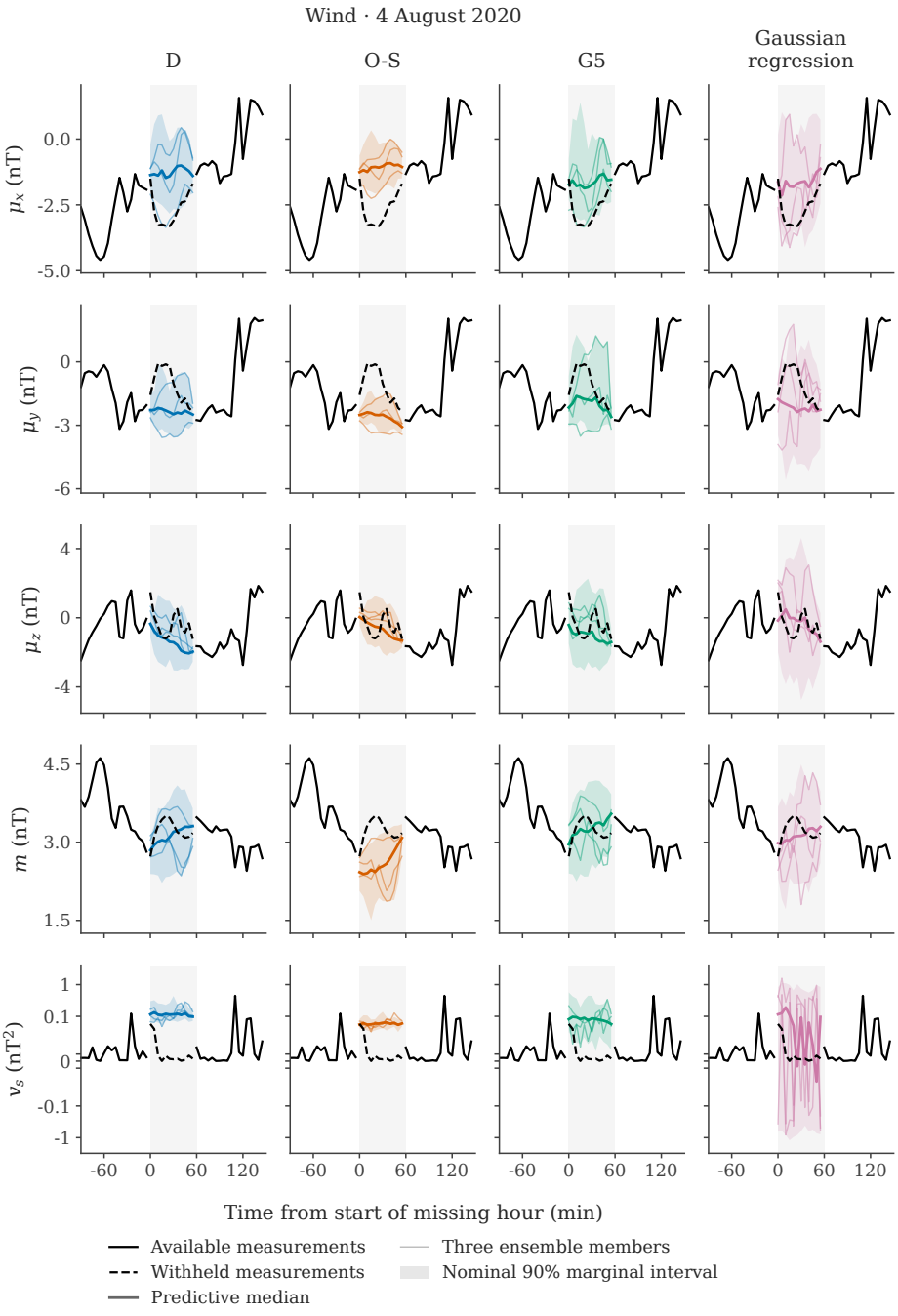



Figure 8: Reconstruction from Wind measurements on 4 August 2020, TEST window 360. The missing hour runs from 18:30 to 19:30 UTC on the OMNI arrival-time grid. Methods, seed choices, units and line styles follow Figure 7. Each method's own ensemble determines its predictive intervals.

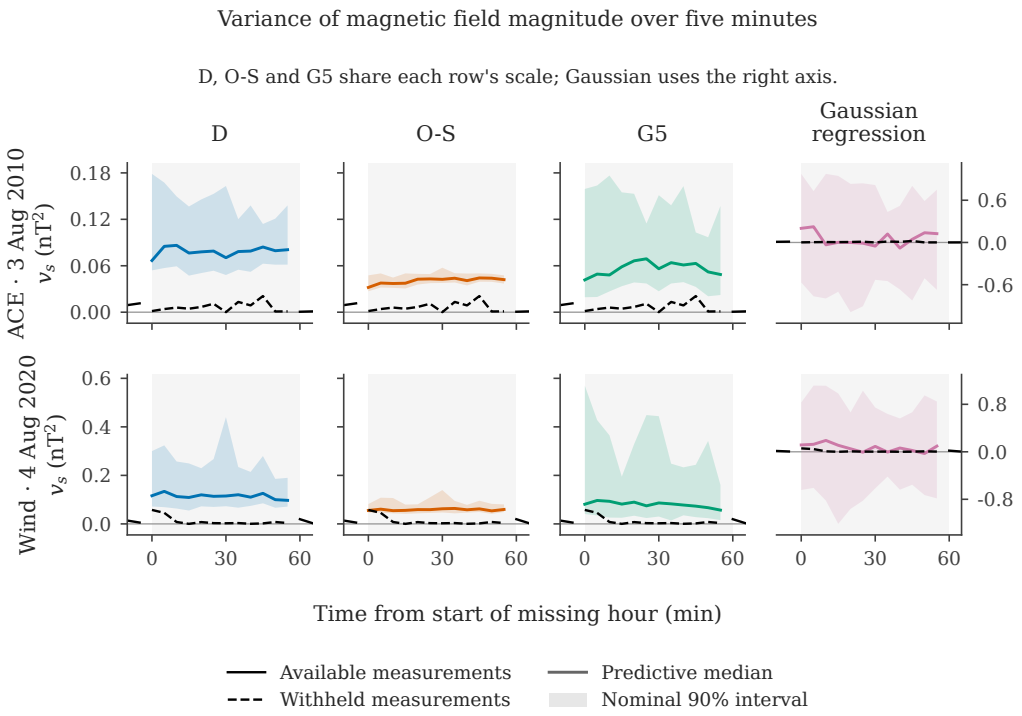



Figure 9: Scalar variance $v_s$ in the same two missing hours. Each value is the variance of the magnitudes of five consecutive vector averages, each over one minute. D, O-S and G5 share a linear scale in each row; Gaussian regression uses a separate linear scale on the right. Vertical distances can be compared directly among the three diffusion methods within a row. Axis limits include every displayed measurement, median and interval endpoint. Solid black lines show available observations, and dashed black lines show withheld measurements. Colored lines show empirical medians, and bands show nominal 90% marginal predictive intervals from all 39 samples. Times mark bin starts relative to the missing hour. The horizontal gray line marks zero; predictions below it are physically impossible because variance is nonnegative. The much wider Gaussian intervals contain all twelve withheld values in each of these two hours.